CADT

Department: Computer Science
Specialization: Software Engineering

Course: Fundamentals Of Data Science

Week-6-Lab: Data Visualization

I. Dataset: House Price Dataset: House Pricing Dataset

Part 1: Univariate Analysis

1. **Task 1: Histogram**
   - Create histogram for:
     - Price
     - Area
   - Questions:
     - Is the distribution normal, skewed left, or skewed right?
     - Are there extreme values?
2. **Task 2: Boxplot**
   - Create boxplot for:
     - Price
     - Area
   - Questions:
     - Explain the graph based on what you see
3. **Task 3: Bar Chart**
   - Create bar chart:
     - Average price by number of bedrooms
   - Questions:
     - Does price increase with bedrooms?

- Is it always increasing?

Part 2: Bivariate & Multivariate

4. **Task 4: Scatter Plot**
   - Plot:
     - Area vs Price
     - Add regression line (Seaborn).
   - Questions:
     - Is the relationship positive or negative?
     - Strong or weak correlation?
     - Any unusual points?

5. **Task 5: Heatmap**

```
# Remove unnecessary columns
df_clean = df.drop(columns=['id', 'long', 'lat', 'zipcode'])
# Select only numeric columns
numeric_df = df_clean.select_dtypes(include=['int64', 'float64'])
# Compute correlation
corr = numeric_df.corr()
# Plot heatmap
plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap (Selected Features Only)")
plt.show()
```

   - Questions:
     - Which feature is most correlated with price?
     - Which features are highly correlated with each other?

II. Dataset: data (send in group)

6. **Task 6: Understand Poverty Situation**
   - Analyze how poverty is distributed across provinces.

- Questions:
    - Are poverty rates evenly distributed?
    - Are there extreme provinces?
    - What does the distribution shape look like?

7. **Task 7: Provincial Comparison**
    - Compare average income across provinces.
    - Questions:
        - Which provinces are highest and lowest?
        - Is there large inequality?

8. **Task 8: Relationship Exploration**
    - Investigate the relationship between poverty rate and average income.
    - Questions:
        - Is there a relationship?
        - Is it positive or negative?
        - Strong or weak?
        - Calculate the correlation between income and poverty

9. **Task 9: Education & Poverty**
    - Explore how literacy rate relates to poverty rate.
    - Questions:
        - Do provinces with higher literacy have lower poverty?
        - Any unusual provinces?

10. **Task 10: Economic Trend**
    - Analyze tourism revenue trends over time for at least 3 provinces.
    - Questions:
        - Which province grows fastest?
        - Which is unstable?
        - What happened during specific years?

11.Task 11: Correlation Matrix

```
corr_matrix =
df[['avg_income_usd','literacy_rate','unemployment_rate','tourism_revenue_million','school_enrol
lment_rate']].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Matrix of Independent Variables")
plt.show()
```

- Questions:
  - Which two variables have the strongest positive correlation?
  - Which variables show a negative relationship with average income
  - Is tourism revenue strongly related to unemployment rate?
  - How does unemployment relate to education indicators?
  - Does higher income strongly correspond to better education outcomes?